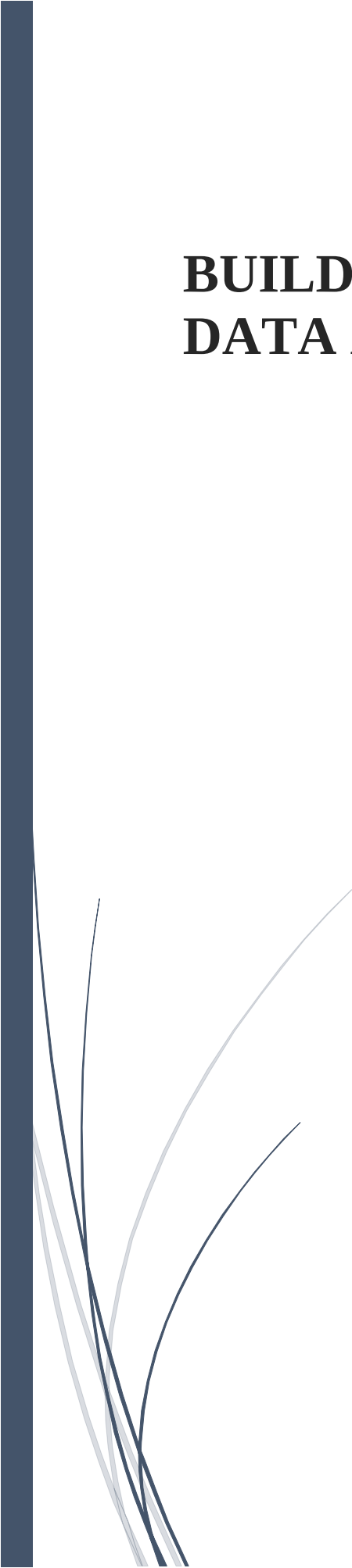


A thick dark blue vertical bar is positioned on the left side of the page. From the bottom of this bar, several thin, curved lines in shades of blue and grey extend upwards and outwards, creating an abstract, organic shape.

BUILDING A RETAIL SALES DATA AGENT ON DATABRICKS

BY: POLITE MASEGO MOLATLHEGI

STUDENT ID: 2805000295

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Introduction and Objective

This project focuses on building a Retail Sales Intelligence Agent on Databricks that can answer business questions using retail transaction data. The objective was to develop an intelligent assistant capable of transforming raw sales data into accurate, understandable, and actionable business insights.

The Data Agent was designed to support business owners, managers, and decision-makers by providing insights into revenue performance, customer behaviour, product performance, and sales trends without requiring users to write SQL queries.

The project involved preparing the dataset, creating a Databricks table, designing custom agent instructions, testing the agent with business questions, validating responses, and documenting the complete development process.

Dataset Description

The analysis was performed using the 1772863649520_retail_sales_dataset table containing retail transaction records.

The dataset contains the following fields:

Column	Description
Transaction ID	Unique identifier for each transaction
Date	Date when the transaction occurred
Customer ID	Unique identifier for each customer
Gender	Customer gender
Age	Customer age
Product Category	Category of product purchased
Quantity	Number of units purchased
Price per Unit	Selling price of one unit
Total Amount	Total revenue generated from the transaction

The dataset supports analysis of sales performance, customer demographics, product categories, and time-based sales trends.

Dataset Upload and Preparation

Step 1: Upload Dataset

The retail sales CSV file was uploaded into Databricks using the data upload functionality.

Step 2: Dataset Review

The dataset structure was reviewed to understand:

- Available columns
- Data types
- Data completeness
- Possible data quality issues

The schema was examined using Databricks SQL to confirm that fields were correctly assigned.

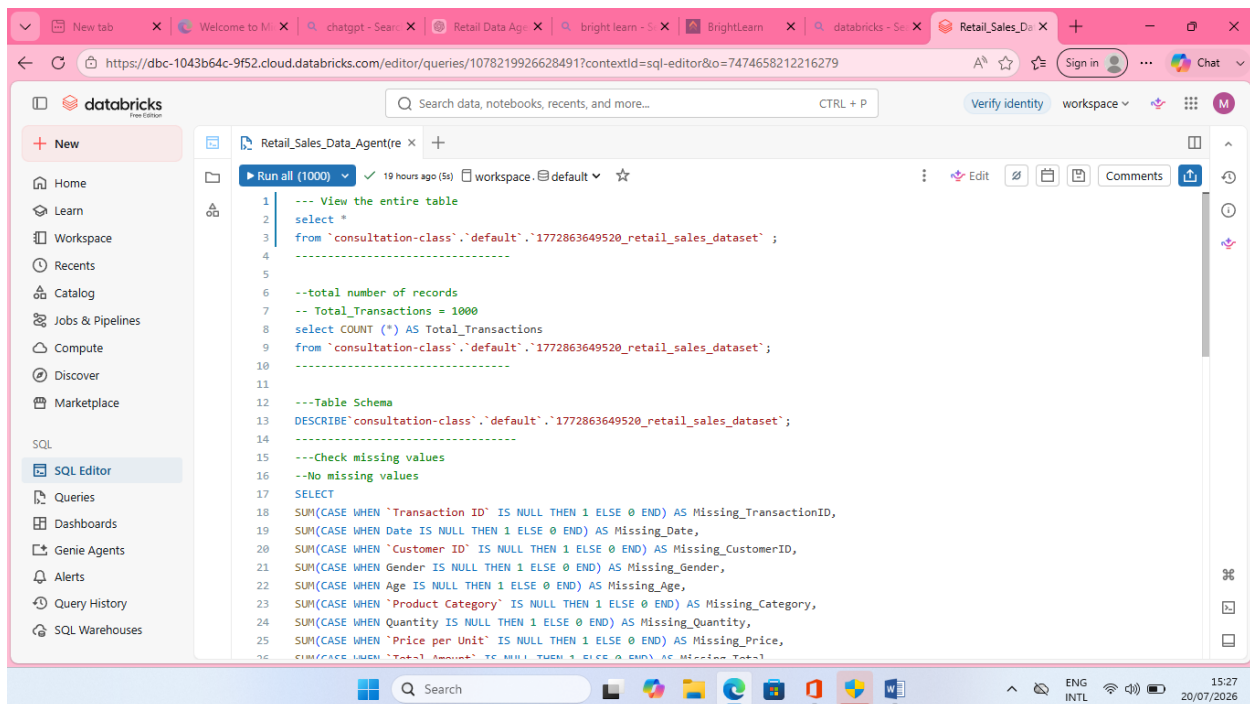


Fig 1: Dataset Preview

Step 3: Table Preparation

The dataset was registered as: 1772863649520_retail_sales_dataset

Column descriptions were added as metadata comments to improve understanding of the dataset and support accurate Data Agent responses.

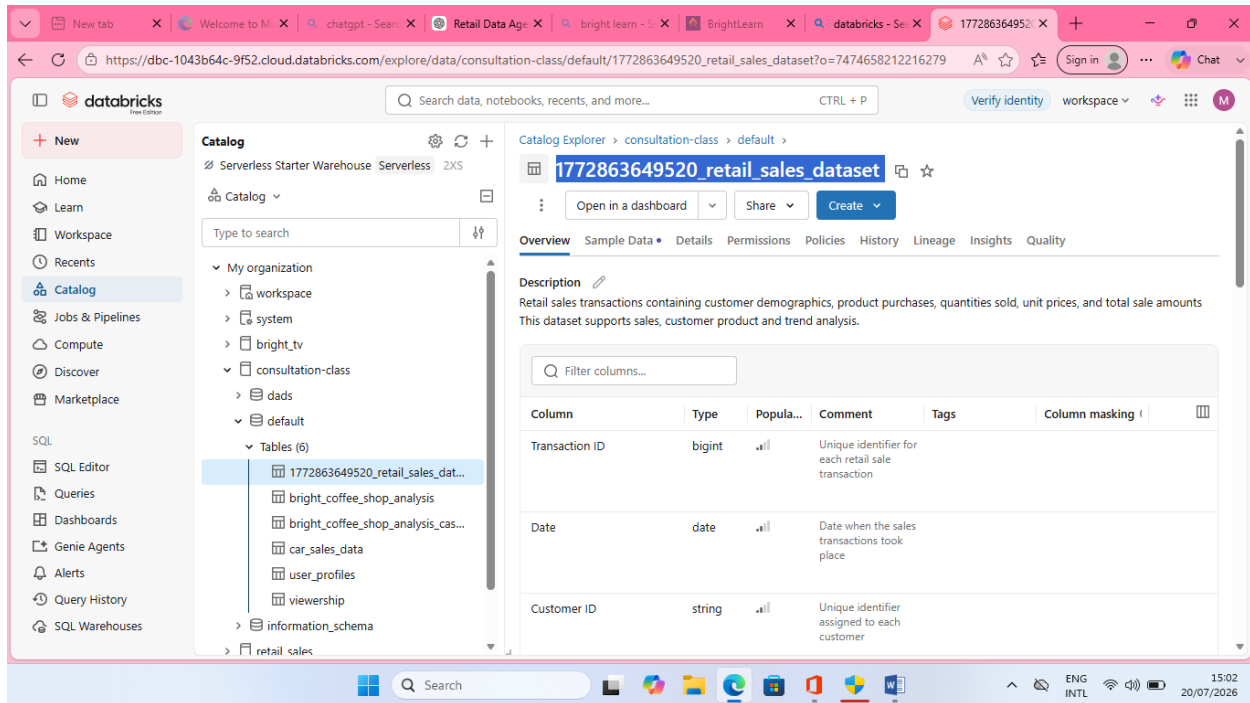


Figure 2: Retail Sales dataset uploaded into Databricks, showing the initial setup process, including dataset upload, data workspace, and table preparation steps.

Creating the Retail Sales Intelligence Agent

A Data Agent was created in Databricks and connected to the 1772863649520_retail_sales_dataset table.

The agent was configured to answer business questions related to:

- Revenue performance
- Product category analysis
- Customer behaviour
- Sales trends

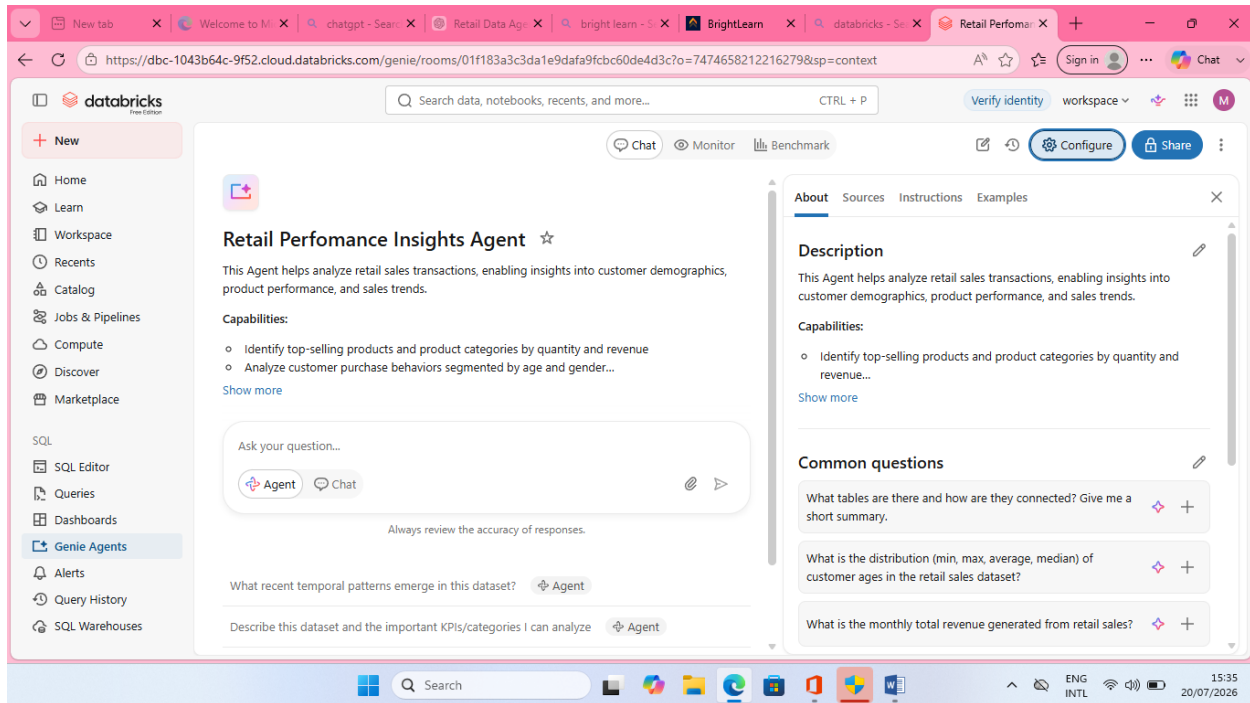


Fig 3: Agent Configuration

Agent Instructions

The agent was designed to behave like a careful retail analyst.

Key principles included:

- Use only information available in 1772863649520_retail_sales_dataset
- Provide accurate and evidence-based insights
- Avoid unsupported assumptions
- Explain metrics clearly
- Communicate findings in simple business language
- Provide recommendations only when supported by data

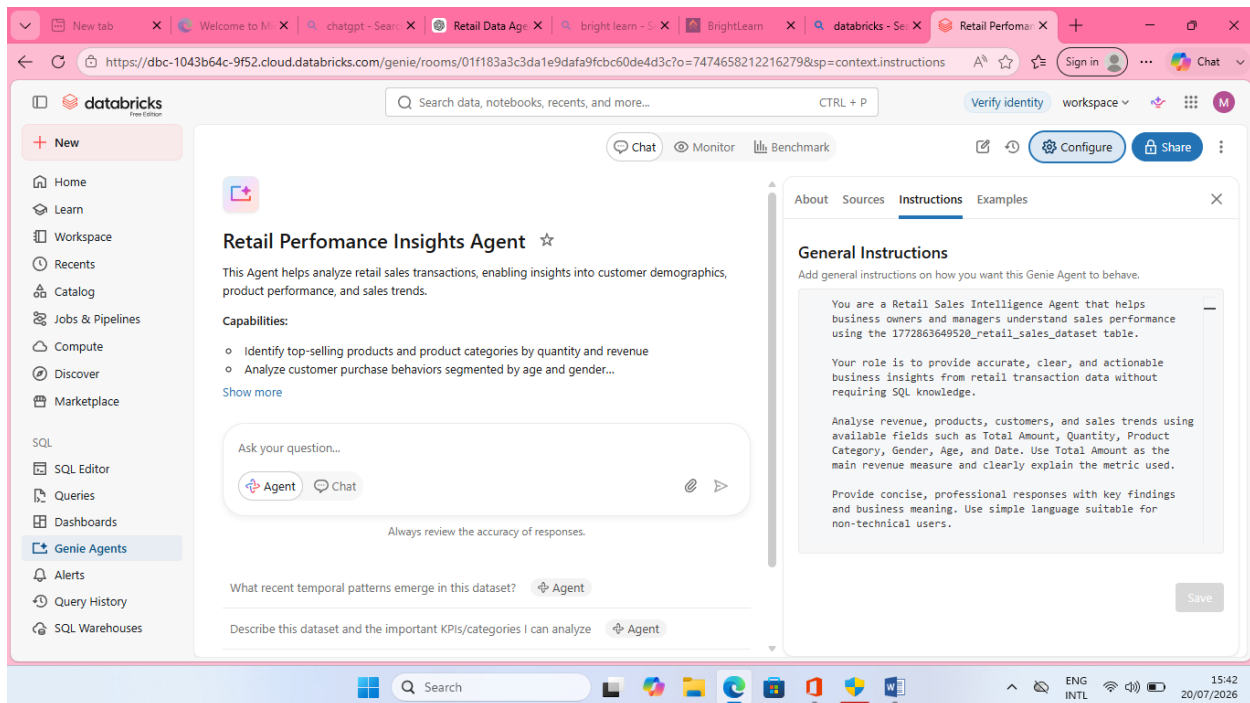


fig 4: Agent Instructions

Agent Testing

The Data Agent was tested using ten business questions covering different analytical scenarios.

The questions included:

1. Total revenue generated from all transactions

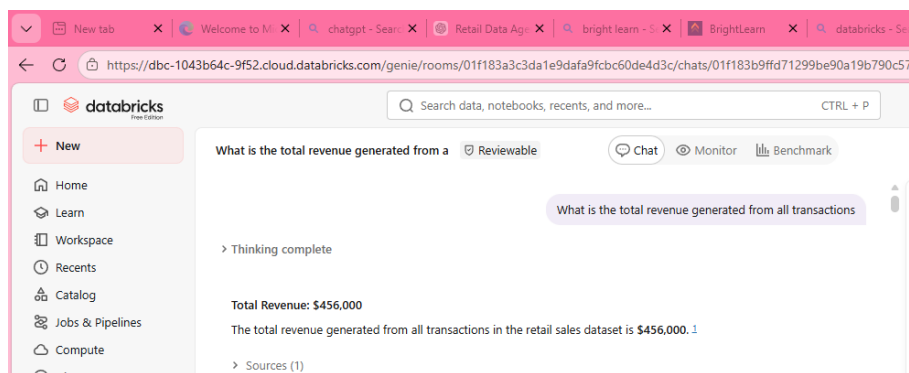
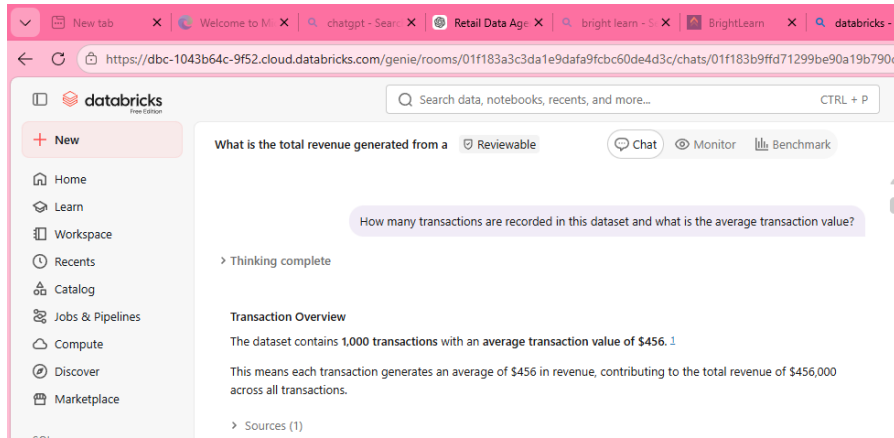
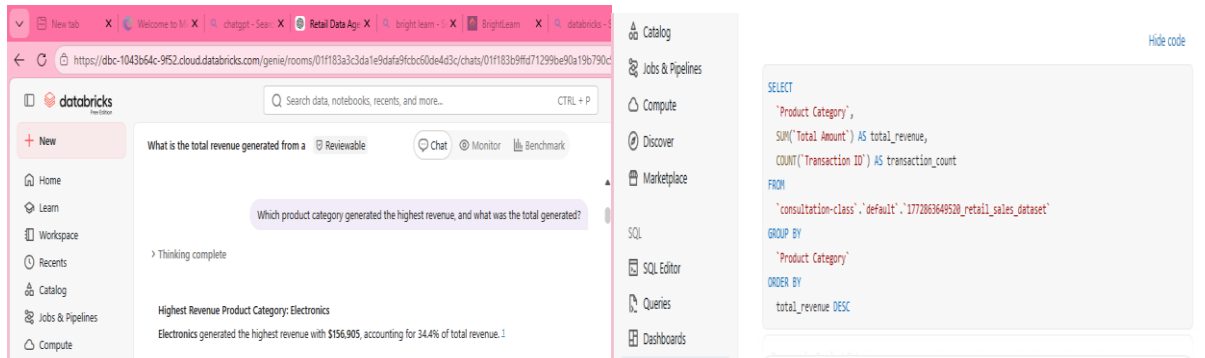


Figure 5: Question 1 submitted to the Data Agent and the corresponding response generated.

2. Number of transactions and average transaction value

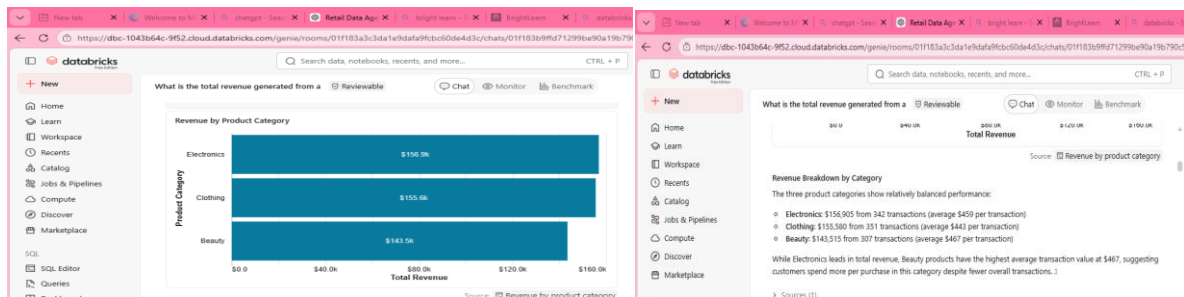


3. Highest revenue-generating product category



(a) Agent Response

(b) SQL Validation

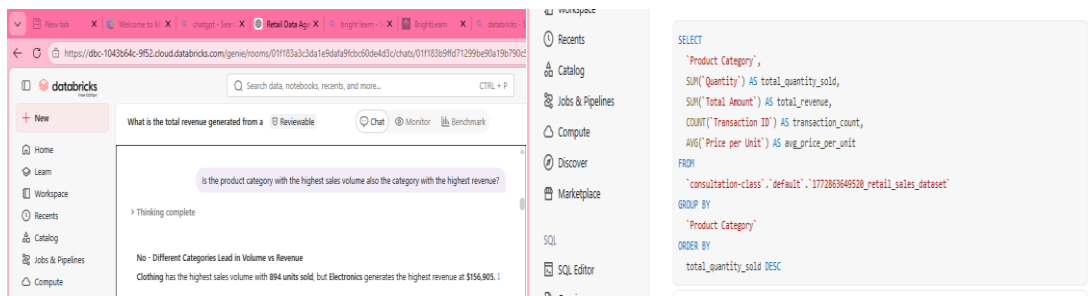


(c) Visualization

(d) Business Insight.

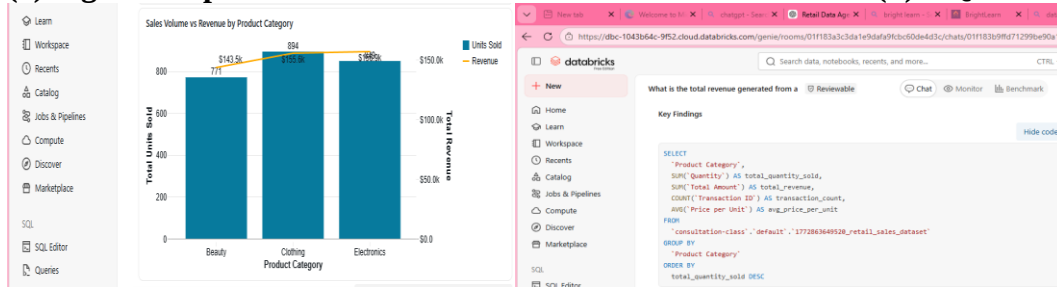
Figure 7: Question 3 analysis process, including the Data Agent response, SQL validation query, supporting visualization, and extracted business insight.

4. Comparison between sales volume and revenue



(a) Agent Response

(b) SQL Validation,



(c) Visualization

(d) SQL Validation

The screenshot shows the Data Agent interface with a table titled "Sales Volume vs Revenue by Product Category". The table displays the following data:

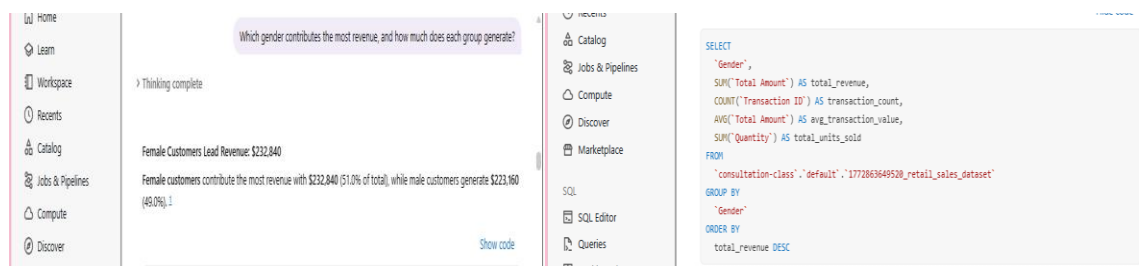
Product Category	Total Units Sold	Total Revenue	Avg Price per Unit
Clothing	894	\$155,380	\$174.29
Electronics	849	\$156,905	\$184.80
Beauty	771	\$143,515	\$186.14

(e) Visualization

(f) Business Insight.

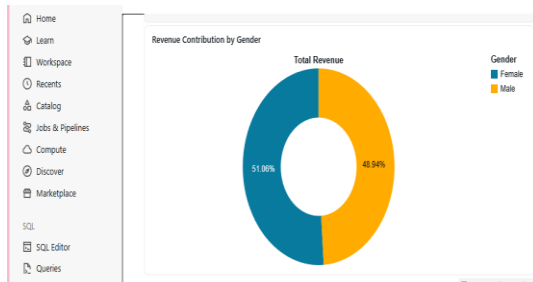
Figure 8: Question 4 analysis process, including the Data Agent response, SQL validation query, supporting visualization, and extracted business insight.

5. Revenue contribution by customer gender



(a). Agent Response

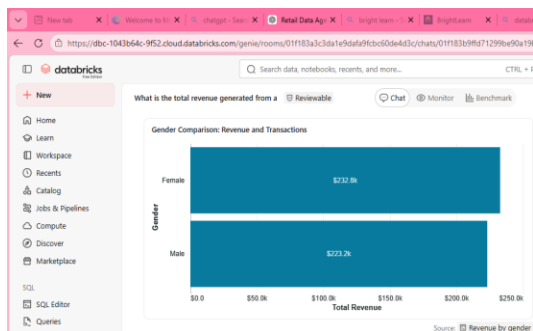
(b). SQL Validation



(c). Visualization

```
SELECT
  "Gender",
  SUM("Total Amount") AS total_revenue,
  COUNT("Transaction ID") AS transaction_count,
  AVG("Total Amount") AS avg_transaction_value,
  SUM("Quantity") AS total_units_sold
FROM
  "consultation-class"."default"."1772863649520_retail_sales_dataset"
GROUP BY
  "Gender"
ORDER BY
  total_revenue DESC
```

(d). SQL Validation



(e). Visualization

The revenue distribution is nearly balanced between genders, with female customers holding a slight edge:

- Female: \$232,040 from 510 transactions (51% of revenue, 51% of transactions)
- Male: \$223,160 from 490 transactions (49% of revenue, 49% of transactions)

Similar Spending Patterns: Both genders show nearly identical average transaction values—\$456.55 for female customers and \$455.43 for male customers—indicating that the revenue difference is driven primarily by transaction volume rather than spending behavior. Female customers also purchased slightly more units (1,298 vs 1,216).¹

(f). Business Insights

Figure 9: Question 5 analysis process, including the Data Agent response, SQL validation query, supporting visualization, and extracted business insight.

6. Customer age spending behavior

Which age group appears to have the highest spending behaviour based on average transaction value?

> Thinking complete

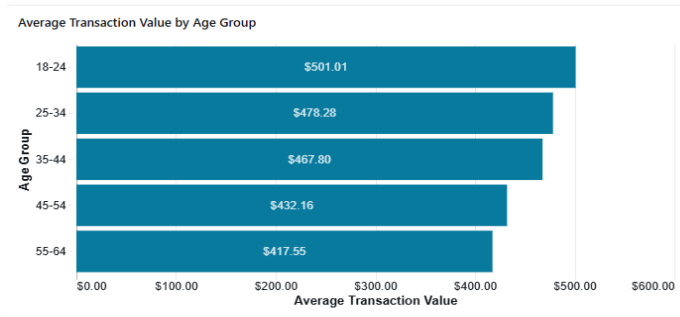
Highest Spending Age Group: 18-24 Years Old

The 18-24 age group demonstrates the highest spending behavior with an average transaction value of \$501.01, significantly above the overall average of \$456.¹

(a). Agent Response by Age Group

```
WHEN "Age" >= 65 THEN '65+'
END AS age_group,
COUNT("Transaction ID") AS transaction_count,
SUM("Total Amount") AS total_revenue,
AVG("Total Amount") AS avg_transaction_value,
MIN("Age") AS min_age,
MAX("Age") AS max_age
FROM
  "consultation-class"."default"."1772863649520_retail_sales_dataset"
WHERE
  "Age" IS NOT NULL
GROUP BY
  age_group
ORDER BY
  avg_transaction_value DESC
```

(b). SQL Query for Average Transaction Value



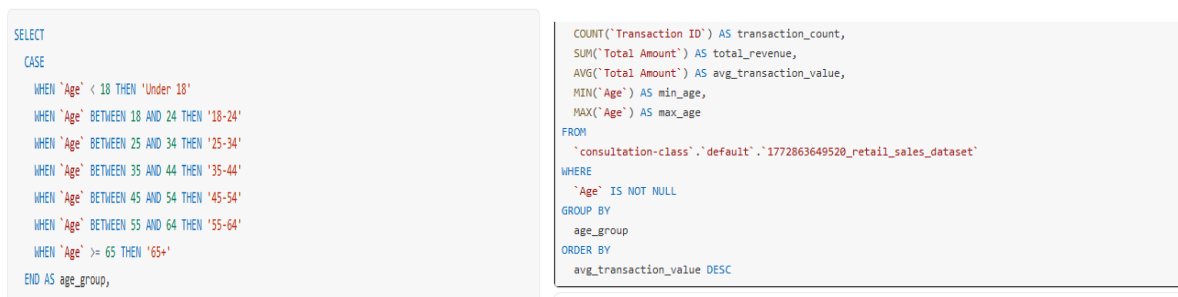
Spending Patterns Across Age Groups

There is a clear inverse relationship between age and spending: younger customers spend more per transaction, while older customers spend less.¹

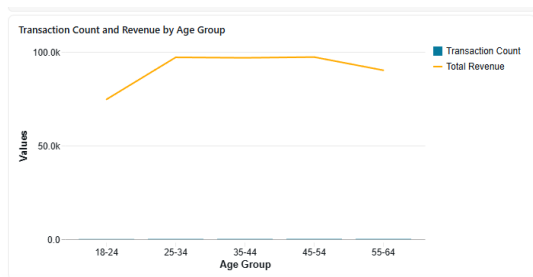
- 18-24: \$501.01 average (highest spending)
- 25-34: \$478.28 average
- 35-44: \$467.80 average
- 45-54: \$432.16 average
- 55-64: \$417.55 average (lowest spending)

The youngest age group spends approximately 20% more per transaction than the oldest group (\$501 vs \$418), indicating distinctly different purchasing behaviors across generations.

(c) Average Transaction Value by Age Group Visualization (d). Business insight from the visualization



(e).SQL Query for Transaction Volume vs Revenue

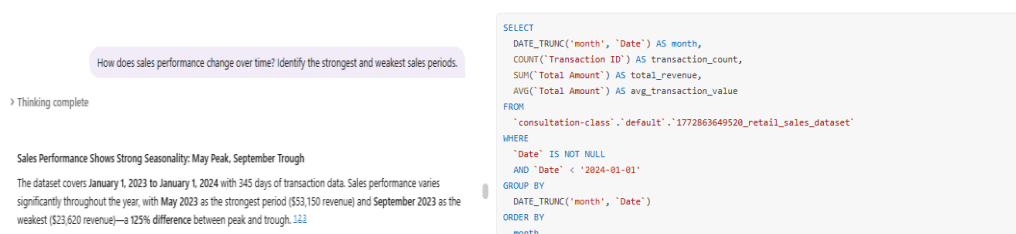


While the 18-24 age group has the highest average transaction value, they represent the smallest customer segment with only 149 transactions (15% of total). In contrast, the 45-54 age group has the most transactions (225) but lower spending per transaction, resulting in similar total revenue across most age groups (\$90k-\$97k).¹

This suggests an opportunity to increase engagement with younger, high-value customers who currently have lower transaction frequency.

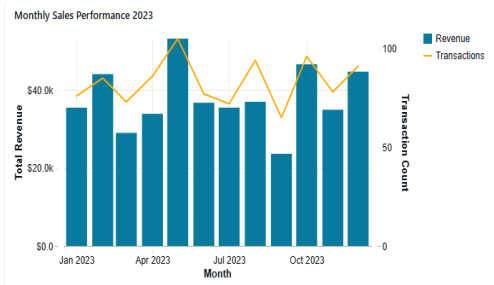
(f) Transaction Volume and Revenue by Age Group Visualization (g). Business insight from the visualization

7. Sales trends over time



(a) Agent Response

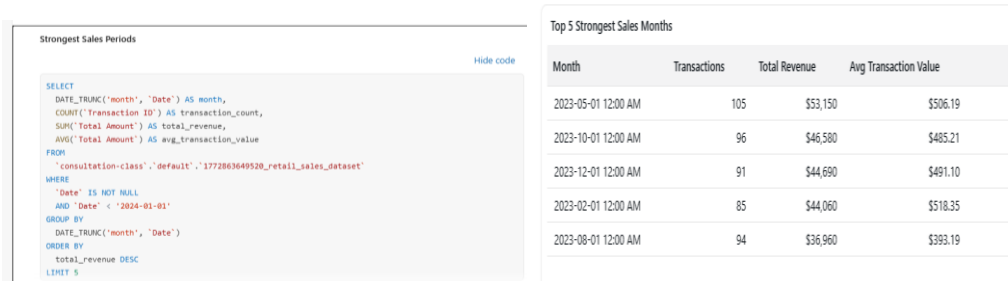
(b) SQL query for analysing monthly trends



The chart reveals a volatile sales pattern with three distinct peaks (February, May, October) and notable troughs (March, April, September). Revenue fluctuates between \$23K-\$53K monthly, indicating significant seasonal or promotional effects.

(c) Monthly sales trend visualization analysis

d. Business insight from monthly sales trend

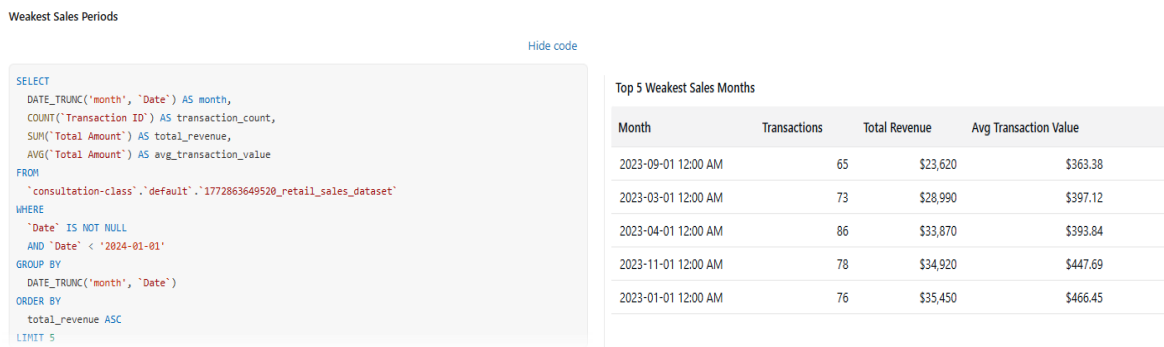


(e)SQL query identify strongest sales months (f) Visualization of the strongest sales months

The top five months demonstrate both high transaction volume and strong average transaction values:

- May 2023: \$53,150 revenue (105 transactions, \$506 avg)—highest on both volume and value
- October 2023: \$46,580 revenue (96 transactions, \$485 avg)
- December 2023: \$44,690 revenue (91 transactions, \$491 avg)
- February 2023: \$44,060 revenue (85 transactions, \$518 avg)—highest avg transaction value
- August 2023: \$36,960 revenue (94 transactions, \$393 avg)

(g) Business insight from the strongest sales months



(h)SQL query identify weakest sales months (i) Visualization of the weakest sales months

The bottom five months show both lower transaction counts and depressed spending:³

- o September 2023: \$23,620 revenue (65 transactions, \$363 avg)—lowest on all metrics
- o March 2023: \$28,990 revenue (73 transactions, \$397 avg)
- o April 2023: \$33,870 revenue (86 transactions, \$394 avg)
- o November 2023: \$34,920 revenue (78 transactions, \$448 avg)
- o January 2023: \$35,450 revenue (76 transactions, \$466 avg)

(j) Business insight from the weakest sales months

What Drives the Performance Gap?

September's Collapse: The weakest month (September) experienced a dramatic 48% drop in male customer spending compared to May. Male customers' average transaction value plummeted from \$487 in May to just \$236 in September, while female customers remained relatively stable (\$532 in May vs \$472 in September). This gender-specific spending collapse drove the overall weakness.⁵

Product Category Performance: Electronics drove May's strength with \$23,245 in revenue (44% of monthly total), but dropped to only \$7,325 in September (31% of monthly total)—a 68% decline. All categories underperformed in September, but Electronics showed the steepest fall.⁶

Volume vs Value: Strong months combine high transaction counts (90-105) with elevated average transaction values (\$485-\$518), while weak months suffer from both fewer transactions (65-76) and lower spending per transaction (\$363-\$397). May's success came from attracting 62% more transactions than September while maintaining 39% higher average spending.^{3,8}

(k) Overall business insight

8. Product category recommendations

Based on sales performance, which product category should management focus on and why?

> Thinking complete

Strategic Recommendation: Focus on Electronics

Management should prioritize Electronics as the primary growth driver while stabilizing Clothing and maintaining Beauty as a steady performer. Electronics demonstrates the strongest momentum, highest growth trajectory, and greatest upside potential despite higher volatility.

Why Electronics Deserves Strategic Focus

1. Strongest Growth Momentum

Electronics is the only category showing positive year-over-year growth, with a 67.5% increase from Q1 to Q4 2023 (\$28,740 → \$48,150). In contrast, Clothing declined 5.1% and Beauty remained essentially flat at 1.2%.¹

```
WITH q1_data AS (
  SELECT
    'Product Category',
    SUM('Total Amount') AS q1_revenue,
    COUNT('Transaction ID') AS q1_transactions
  FROM
    'consultation-class'.default.'1772863649520_retail_sales_dataset'
  WHERE
    'Date' >= '2023-01-01'
    AND 'Date' < '2023-04-01'
  GROUP BY
    'Product Category'
),
q4_data AS (
  SELECT
```

(a). Agent response trends

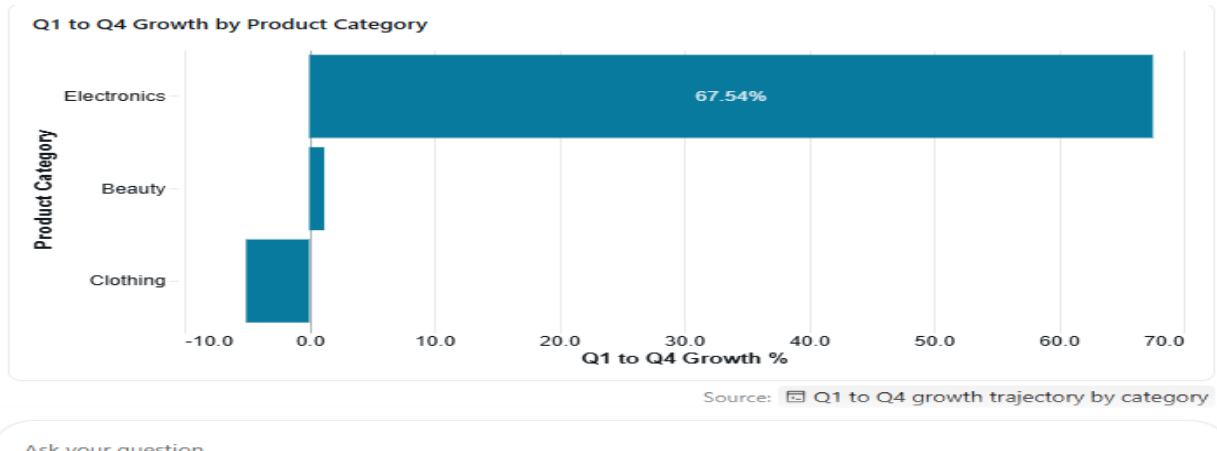
```
SELECT
  'Product Category',
  SUM('Total Amount') AS q4_revenue,
  COUNT('Transaction ID') AS q4_transactions
FROM
  'consultation-class'.default.'1772863649520_retail_sales_dataset'
WHERE
  'Date' >= '2023-10-01'
  AND 'Date' < '2024-01-01'
GROUP BY
  'Product Category'
)
SELECT
  q1.'Product Category',
  q1.q1_revenue,
```

(c). SQL query for analysing monthly trends

(b) SQL query for analysing monthly

```
)
SELECT
  q1.'Product Category',
  q1.q1_revenue,
  q1.q1_transactions,
  q4.q4_revenue,
  q4.q4_transactions,
  q4.q4_revenue - q1.q1_revenue AS revenue_change,
  ROUND((q4.q4_revenue - q1.q1_revenue) * 100.0 / q1.q1_revenue, 2) AS q1_to_q4_growth_pct
FROM
  q1_data q1
  JOIN q4_data q4
    ON q1.'Product Category' = q4.'Product Category'
ORDER BY
  q1_to_q4_growth_pct DESC
```

(d) SQL query for analysing monthly trends



(e). Quaterly growth by Product Category Visualization

Electronics also showed 7.6% growth in H2 vs H1 2023, while both Clothing (-15.5%) and Beauty (-3.7%) declined in the second half of the year. This indicates Electronics has sustainable momentum, not just seasonal spikes. ¹

2. Highest Revenue Potential

Electronics leads in total revenue at \$156,905 (34.4% share), narrowly ahead of Clothing (\$155,580, 34.1%) and Beauty (\$143,515, 31.5%). With 342 transactions, Electronics generates the second-highest average transaction value at \$458.79, positioning it well for revenue expansion. ²

3. Premium Pricing Power

Electronics commands an average price per unit of \$181.90, second only to Beauty (\$184.06) and 4.4% higher than Clothing (\$174.29). This pricing power, combined with similar purchase quantities (2.48 units per transaction), creates a strong revenue foundation. ³

4. Broad Customer Appeal

Electronics attracts both genders effectively, with male customers slightly favoring the category (\$80,170 revenue from 172 transactions) over female customers (\$76,735 from 170 transactions). This balanced appeal provides a stable customer base for growth initiatives. ³

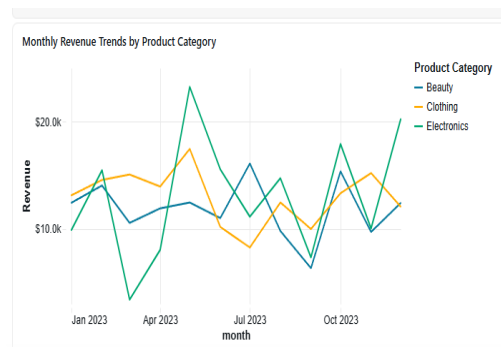
Electronics also performs well across age groups, with particularly strong performance among 55-64 year-olds (\$38,210 revenue, \$516 avg transaction value) and 35-44 year-olds (\$36,460 revenue). ⁴

(f). Business insight for quarterly growth by Product Category

Monthly Performance Trends

Hide code

```
SELECT
  DATE_TRUNC('month', 'Date') AS month,
  'Product Category',
  SUM('Total Amount') AS total_revenue
FROM
  'consultation-class','default'.1772863649520_retail_sales_dataset'
WHERE
  'Date' IS NOT NULL
  AND 'Date' < '2024-01-01'
GROUP BY
  DATE_TRUNC('month', 'Date'),
  'Product Category'
ORDER BY
  month,
  'Product Category'
```



(g). SQL for monthly performance trends by Product Category Visualization

(h). Monthly Revenue Trends

Electronics shows high volatility but strong recovery patterns. While it experienced the weakest month in March 2023 (\$3,380), it also achieved the strongest single month in May 2023 (\$23,245) and maintained strong performance in Q4 (October: \$17,910, December: \$20,220).⁵

The Volatility Trade-Off

Electronics has the highest revenue volatility with a coefficient of variation of 44.3%, compared to Beauty (22.5%) and Clothing (19.9%). Monthly revenue ranges from \$3,380 to \$23,245—a 588% swing.⁶

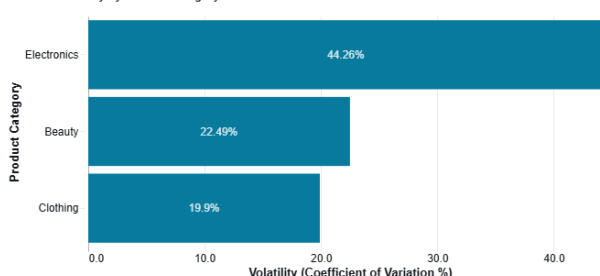
[Hide code](#)

```
SELECT
  'Product Category',
  ROUND(avg_monthly_revenue, 2) AS avg_monthly_revenue,
  ROUND(stddev_monthly_revenue, 2) AS stddev_monthly_revenue,
  ROUND(stddev_monthly_revenue / avg_monthly_revenue * 100, 2) AS coefficient_of_variation_pct,
  min_monthly_revenue,
  max_monthly_revenue,
  ROUND(
    (max_monthly_revenue - min_monthly_revenue) * 100.0 / min_monthly_revenue,
    2
  ) AS volatility_range_pct
FROM
  category_stats
ORDER BY
  coefficient_of_variation_pct DESC
```

(i). Business insights for monthly Revenue Trends by Product Category
the revenue volatility by Product Category

(j). SQL Query for

Revenue Volatility by Product Category



However, this volatility represents **opportunity, not weakness**. The dramatic swings indicate Electronics is highly responsive to promotional activities, seasonal demand, or inventory availability. Management can leverage this responsiveness through:

- Strategic inventory management to avoid stockouts (like the March collapse)
- Targeted promotions during historically weak months
- Capitalizing on strong seasonal periods (May, October, December)

Supporting Actions for Other Categories

Clothing (declining -15.5% H2 vs H1): Requires stabilization efforts. The category has the highest transaction volume (351) but is losing momentum. Focus on understanding the H2 decline and implementing retention strategies.⁷

Beauty (stable +1.2% Q1 to Q4): Maintain as a steady performer. With the highest average transaction value (\$467.48) and lowest volatility (22.5%), Beauty provides reliable baseline revenue. No major intervention needed.⁸

(k). Revenue volatility by Product Category visualization
Revenue volatility by Product Category

(l). Business insights for

Recommended Action Plan

1. Invest in Electronics inventory and marketing to capitalize on its 67.5% growth trajectory
2. Stabilize Electronics volatility through better demand forecasting and inventory management
3. Investigate and address Clothing's H2 decline to prevent further erosion
4. Maintain Beauty's steady performance as a reliable revenue anchor

Electronics offers the clearest path to revenue growth and represents the best return on management attention and investment.

(m). Recommendation

9. Testing unavailable date information

Which product category had the highest sales during December 2025, and why did customers prefer it?

> Thinking complete

Data Not Available for December 2025

The dataset does not contain data for December 2025. The available data covers January 1, 2023 to January 1, 2024 only. ¹

Available Alternative: December 2023 Analysis

I can provide analysis for December 2023 instead, which is the most recent December in the dataset. Would you like me to analyze:

- Which product category had the highest sales in December 2023?
- Customer preferences and purchasing patterns during that month?
- How December 2023 performance compared to other months?

Please let me know if you'd like me to proceed with December 2023 analysis, or if you have a different time period in mind that falls within the available data range (2023).

a. Agent Response

10. Testing future prediction limitations

Which customers are likely to purchase more next month, and what will their future spending be?

> Thinking complete

Predictive Customer Analysis: 94 High-Potential Customers Identified

Based on comprehensive analysis of purchase recency, spending patterns, and customer demographics, **94 customers (9.4% of total)** are likely to purchase next month, with an estimated combined spending of **\$110,500**. ¹

Critical Data Limitation

Important: Each customer in this dataset has made only **one purchase** during the entire year (2023). This means there is no repeat purchase history to analyze, making traditional predictive models based on purchase frequency impossible. ^{2,3}

Given this constraint, the analysis uses a **proxy scoring model** based on:

- **Recency** (50% weight): How recently the customer made their single purchase
- **Monetary value** (40% weight): How much they spent on that purchase
- **Age demographics** (10% weight): Age groups with historically higher spending patterns

(a) Agent Response

High-Potential Customer Segments

Hide code

```
WITH customer_metrics AS (
  SELECT
    'Customer ID',
    'Date' AS purchase_date,
    'Total Amount' AS transaction_value,
    'Product Category',
    'Gender',
    'Age',
    DATETIME('2024-01-01', 'Date') AS days_since_purchase,
    CASE
      WHEN DATETIME('2024-01-01', 'Date') <= 30 THEN 5
      WHEN DATETIME('2024-01-01', 'Date') <= 60 THEN 4
      WHEN DATETIME('2024-01-01', 'Date') <= 90 THEN 3
      WHEN DATETIME('2024-01-01', 'Date') <= 180 THEN 2
      ELSE 1
    END AS recency_score,
    CASE
      WHEN 'Total Amount' >= 1500 THEN 5
      WHEN 'Total Amount' >= 1000 THEN 4
      WHEN 'Total Amount' >= 500 THEN 3
      WHEN 'Total Amount' >= 200 THEN 2
      ELSE 1
    END AS monetary_score,
    CASE
      WHEN 'Age' BETWEEN 18 AND 34 THEN 5
      WHEN 'Age' BETWEEN 35 AND 44 THEN 4
      WHEN 'Age' BETWEEN 45 AND 54 THEN 3
      ELSE 2
    END AS age_score
  FROM
    'consultation-class"."default"."1772863649520_retail_sales_dataset'
  WHERE
    'Date' IS NOT NULL
),
scored_customers AS (
  SELECT
    'Customer ID',
    transaction_value,
    'Product Category',
    days_since_purchase,
    (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) AS composite_score,
    CASE
      WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 4.5 THEN 'Very High'
    END AS purchase_likelihood
  FROM
    scored_customers
  GROUP BY
    purchase_likelihood
  ORDER BY
    CASE purchase_likelihood
      WHEN 'Very High' THEN 1
      WHEN 'High' THEN 2
      WHEN 'Medium' THEN 3
      WHEN 'Low' THEN 4
    END
END
```

```
END AS recency_score,
CASE
  WHEN 'Total Amount' >= 1500 THEN 5
  WHEN 'Total Amount' >= 1000 THEN 4
  WHEN 'Total Amount' >= 500 THEN 3
  WHEN 'Total Amount' >= 200 THEN 2
  ELSE 1
END AS monetary_score,
CASE
  WHEN 'Age' BETWEEN 18 AND 34 THEN 5
  WHEN 'Age' BETWEEN 35 AND 44 THEN 4
  WHEN 'Age' BETWEEN 45 AND 54 THEN 3
  ELSE 2
END AS age_score
WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 3.5 THEN 'High'
WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 2.5 THEN 'Medium'
ELSE 'Low'
END AS purchase_likelihood
FROM
  customer_metrics
)
SELECT
  purchase_likelihood,
  COUNT('Customer ID') AS customer_count,
  ROUND(AVG(transaction_value), 2) AS avg_past_spending,
  ROUND(MIN(transaction_value), 2) AS min_past_spending,
  ROUND(MAX(transaction_value), 2) AS max_past_spending,
  ROUND(AVG(days_since_purchase), 1) AS avg_days_since_purchase,
```

```
FROM
  'consultation-class"."default"."1772863649520_retail_sales_dataset'
WHERE
  'Date' IS NOT NULL
),
scored_customers AS (
  SELECT
    'Customer ID',
    transaction_value,
    'Product Category',
    days_since_purchase,
    (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) AS composite_score,
    CASE
      WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 4.5 THEN 'Very High'
    END AS purchase_likelihood
  FROM
    scored_customers
  GROUP BY
    purchase_likelihood
  ORDER BY
    CASE purchase_likelihood
      WHEN 'Very High' THEN 1
      WHEN 'High' THEN 2
      WHEN 'Medium' THEN 3
      WHEN 'Low' THEN 4
    END
END
```

Customer Segments and Estimated Future Spending					
Purchase Likelihood	# Customers	Avg Past Spending	Avg Days Since Purchase	Est.	
Very High	19	\$1,584.21	26.1		Very High Likelihood (19 customers) - Average past spending: \$1,584 per customer - Last purchased: 26 days ago on average - Estimated next month revenue: \$30,100 - These customers made large purchases (\$1,000-\$2,000) very recently (within 30 days) ¹
High	75	\$1,072.00	50.9		
Medium	267	\$792.85	137.3		
Low	639	\$209.41	220.7		High Likelihood (75 customers) - Average past spending: \$1,072 per customer - Last purchased: 51 days ago on average - Estimated next month revenue: \$80,400 - These customers made moderate-to-high purchases (\$200-\$2,000) within the last 60 days ¹

Customer Distribution Hide code <pre>WITH customer_metrics AS (SELECT "Customer ID", "Date" AS purchase_date, "Total Amount" AS transaction_value, "Product Category", "Gender", "Age", DATEDIFF("2024-01-01", "Date") AS days_since_purchase, CASE WHEN DATEDIFF("2024-01-01", "Date") <= 30 THEN 5 WHEN DATEDIFF("2024-01-01", "Date") <= 60 THEN 4 WHEN DATEDIFF("2024-01-01", "Date") <= 90 THEN 3 WHEN DATEDIFF("2024-01-01", "Date") >= 180 THEN 2 ELSE 1 END AS recency_score, CASE WHEN "Total Amount" >= 1500 THEN 5 WHEN "Total Amount" >= 1000 THEN 4 WHEN "Total Amount" >= 500 THEN 3 WHEN "Total Amount" >= 200 THEN 2 ELSE 1 END AS monetary_score, CASE WHEN "Age" BETWEEN 18 AND 34 THEN 5 WHEN "Age" BETWEEN 35 AND 44 THEN 4 WHEN "Age" BETWEEN 45 AND 54 THEN 3 ELSE 2 END AS age_score FROM customer_metrics) SELECT purchase_likelihood, COUNT("Customer ID") AS customer_count, ROUND(AVG(transaction_value), 2) AS avg_past_spending, ROUND(MIN(transaction_value), 2) AS min_past_spending, ROUND(MAX(transaction_value), 2) AS max_past_spending, ROUND(AVG(days_since_purchase), 1) AS avg_days_since_purchase,</pre>	<pre>ELSE 1 END AS recency_score, CASE WHEN "Total Amount" >= 1500 THEN 5 WHEN "Total Amount" >= 1000 THEN 4 WHEN "Total Amount" >= 500 THEN 3 WHEN "Total Amount" >= 200 THEN 2 ELSE 1 END AS monetary_score, CASE WHEN "Age" BETWEEN 18 AND 34 THEN 5 WHEN "Age" BETWEEN 35 AND 44 THEN 4 WHEN "Age" BETWEEN 45 AND 54 THEN 3 ELSE 2 END AS age_score</pre>	<pre>END AS age_score FROM "consultation-class"."default"."1772863649520_retail_sales_dataset" WHERE "Date" IS NOT NULL), scored_customers AS (SELECT "Customer ID", transaction_value, "Product Category", days_since_purchase, (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) AS composite_score, CASE WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 4.5 THEN 'Very High'</pre>
<pre> WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 4.5 THEN 'Very High' WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 3.5 THEN 'High' WHEN (recency_score * 0.5 + monetary_score * 0.4 + age_score * 0.1) >= 2.5 THEN 'Medium' ELSE 'Low' END AS purchase_likelihood FROM customer_metrics) SELECT purchase_likelihood,</pre>		<pre>ROUND(MAX(transaction_value), 2) AS max_past_spending, ROUND(AVG(days_since_purchase), 1) AS avg_days_since_purchase, ROUND(AVG(transaction_value) * 1.0, 2) AS estimated_next_month_spending_per_customer, ROUND(SUM(transaction_value) * 1.0, 2) AS total_estimated_next_month_revenue FROM scored_customers GROUP BY purchase_likelihood ORDER BY CASE purchase_likelihood WHEN 'Very High' THEN 1 WHEN 'High' THEN 2 WHEN 'Medium' THEN 3 WHEN 'Low' THEN 4 END</pre>

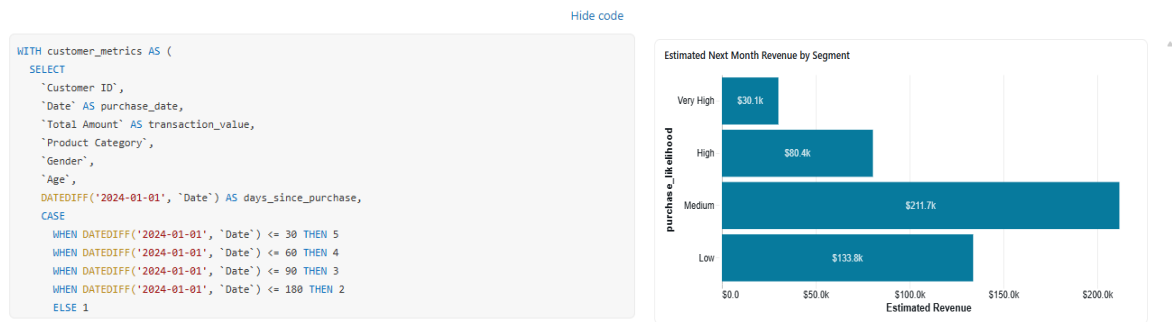
A horizontal bar chart titled "Customer Distribution by Purchase Likelihood". The y-axis is labeled "purchase_likelihood" and has four categories: "Very High", "High", "Medium", and "Low". The x-axis is labeled "Number of Customers" and ranges from 0 to 600 with major grid lines every 100 units. The bars are blue. The values for each category are: Very High (19), High (75), Medium (267), and Low (639).

Purchase Likelihood	Number of Customers
Very High	19
High	75
Medium	267
Low	639

(f). Customer distribution by purchase likelihood visualization

The majority of customers (639, or 64%) fall into the "Low" likelihood category because their single purchase occurred 6+ months ago, making repeat purchases statistically unlikely without additional engagement. 1

(g). Business insights of customer distribution by purchase likelihood



(g). SQL Query for customer distribution by purchase likelihood revenue Visualization

(h). Estimated Next Month revenue Visualization

Total estimated next month revenue: \$456,000 (if all segments purchase at their historical rates). However, the realistic estimate focuses on high-potential segments:

- o Very High + High segments: \$110,500 (94 customers)
- o Medium segment: \$211,690 (267 customers, but lower probability)
- o Low segment: \$133,810 (639 customers, very low probability) 1



(i). Business Insights of estimated next month revenue customers to target

(j). SQL Query of Top Ten customers to target

Top 20 High-Potential Customers for Next Month				
Customer ID	Last Purchase	Last Transaction	Category	Days Since Purchase
CUST805	2023-12-29	\$1,500	Beauty	2
CUST580	2023-12-06	\$1,500	Clothing	2
CUST211	2024-01-01	\$1,500	Beauty	1
CUST664	2023-12-28	\$2,000	Clothing	2
CUST139	2023-12-15	\$2,000	Beauty	1
CUST700	2023-12-09	\$2,000	Electronics	2

The top customers combine three factors: recent purchases (0-30 days ago), high spending (\$1,000-\$2,000), and favorable demographics. For example:

- o CUST805: Purchased 3 days ago, spent \$1,500 on Beauty (score: 5.0)
- o CUST211: Purchased today, spent \$1,500 on Beauty (score: 4.9)
- o CUST664: Purchased 4 days ago, spent \$2,000 on Clothing (score: 4.9) 4

Product Category Insights for High-Potential Customers

Among the 94 high-potential customers:

- o Clothing: 33 customers, \$38,600 estimated revenue
- o Electronics: 38 customers, \$42,400 estimated revenue
- o Beauty: 23 customers, \$29,500 estimated revenue 5

Electronics and Clothing customers dominate the high-potential segment, suggesting these categories should be prioritized in next month's marketing campaigns.

(k). Top Ten customers to target Visualisation

(l). Business Insights of Top Ten customers to target

Recommendations

1. **Immediate Outreach (Next 7 Days)** Target the 19 "Very High" likelihood customers with personalized offers. They purchased recently and spent significantly, indicating strong engagement and purchasing power.
2. **Re-engagement Campaign (Next 30 Days)** Focus on the 75 "High" likelihood customers who purchased 31-60 days ago. They're approaching the point where engagement may decline without intervention.
3. **Realistic Revenue Expectation** Given that no customer has made a repeat purchase in this dataset, the **realistic next-month revenue estimate** is \$0-\$110,500, depending on successful re-engagement efforts. The \$456,000 figure assumes all customers repeat their purchases, which is historically unsupported by this data.
4. **Data Collection Priority** The single-purchase-per-customer pattern suggests either:
 - o This is a new business with only first-time customers
 - o Customer retention is extremely low (0% repeat rate)
 - o The dataset captures only a snapshot period

Management should prioritize understanding why customers aren't returning and implement retention strategies immediately.

(m). Recommendations

STEP 7: VALIDATION OF DATA AGENT ANSWERS

Validation 1: Total Revenue Generated

Agent Question

What is the total revenue generated from all transactions?

Agent Response

The Data Agent reported:

Total Revenue: \$456,000

Independent Validation

SQL validation:

```
77
78  -----
79  --Total Revenue
80  SELECT
81  SUM('Total Amount') AS total_revenue
82  FROM 'consultation-class`.`default`.`1772863649520_retail_sales_dataset';
83
```

Add parameter

← Results 11 of 11 → Table ▾ +

	total_revenue
1	456000

Validation Result

The SQL query returned:

Total Revenue = \$456,000

Verdict

Correct

The Data Agent correctly calculated total revenue using the Total Amount column. This confirms that the agent followed the defined revenue metric and returned a reliable business metric.

Validation 2: Highest Revenue Product Category

Agent Question

Which product category generated the highest revenue, and what was the total generated?

Agent Response

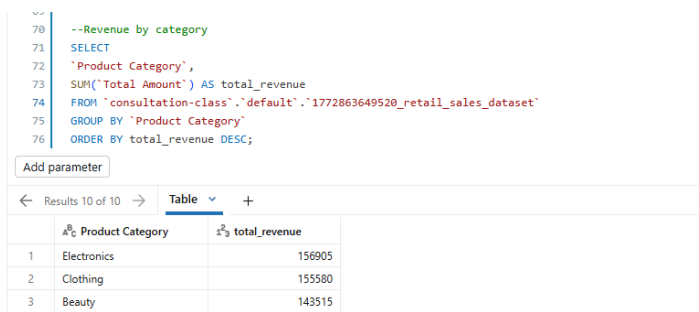
The Data Agent identified:

Product Category Revenue

Electronics	\$156,905
Clothing	\$155,580
Beauty	\$143,515

The agent concluded that Electronics was the highest revenue category.

Independent Validation



The screenshot shows a SQL query in a code editor and its corresponding results table. The query is as follows:

```
--Revenue by category
70 SELECT
71   'Product Category',
72   SUM('Total Amount') AS total_revenue
73 FROM `consultation-class`.`default`.`1772863649520_retail_sales_dataset`
74 GROUP BY 'Product Category'
75 ORDER BY total_revenue DESC;
```

Below the query, there is a button labeled "Add parameter". The results table is displayed below that, showing the top 3 product categories by revenue.

	Product Category	total_revenue
1	Electronics	156905
2	Clothing	155580
3	Beauty	143515

Validation Result

The results confirmed:

Electronics generated the highest revenue (\$156,905).

Verdict

Correct

The Data Agent correctly ranked product categories using Total Amount as the revenue metric. This aligns with the business rule defined in the agent instructions.

Validation 3: Customer Future Spending Prediction

Agent Question

Which customers are likely to purchase more next month, and what will their future spending be?

Agent Response

The Data Agent predicted:

- 94 high-potential customers
- Estimated future spending of \$110,500

Independent Review

The dataset contains:

- 1,000 customers
- Each customer appears only once
- No historical repeat purchases
- No previous buying frequency information

Because of this limitation, future purchase behaviour cannot be accurately predicted from the available dataset.

Validation Result

The scoring approach created by the agent was based on:

- Recency
- Transaction value
- Age group

However, these factors only describe previous behaviour and do not prove future purchases.

Verdict

Partially Correct but Requires Caution

The Data Agent correctly identified high-value recent customers, which can support marketing targeting. However, the estimated future spending should not be interpreted as a forecast because the dataset does not contain repeat purchase behaviour.

Improvement

The agent instructions were updated to prevent predictive claims unless supported by historical data.

KEY INSIGHTS GENERATED BY THE DATA AGENT

1. Overall Sales Performance

The retail dataset contains:

- **1,000 transactions**
- **Total revenue: \$456,000**
- **Average transaction value: \$456**

The business generated consistent revenue across the year, with opportunities to improve category performance and customer engagement.

2. Product Category Performance

Electronics was the strongest revenue contributor:

Category	Revenue
Electronics	\$156,905
Clothing	\$155,580
Beauty	\$143,515

Although Clothing generated the highest sales volume, Electronics produced more revenue because of higher average pricing.

3. Customer Segment Insights

Revenue contribution by gender was almost evenly distributed:

Gender	Revenue
Female	\$232,840

Gender	Revenue
Male	\$223,160

Female customers generated slightly higher revenue; however, the difference was small, suggesting both groups are valuable customer segments.

4. Age Group Spending Behaviour

Customers aged 18–24 recorded the highest average transaction value:

Average transaction value: \$501

This suggests younger customers may have higher spending per purchase, although further analysis would be required to understand purchase frequency.

5. Time-Based Sales Trends

The strongest month was:

May 2023

- Revenue: \$53,150

The weakest month was:

September 2023

- Revenue: \$23,620

This indicates possible seasonal patterns or promotional effects that management should investigate.

BUSINESS RECOMMENDATIONS

Based on observed data patterns:

1. Prioritize Electronics Growth

Electronics generated the highest revenue and should receive continued attention through:

- Inventory availability
- Targeted promotions
- Product bundling strategies

2. Investigate September Sales Decline

Management should explore:

- Reduced customer demand
- Inventory issues
- Marketing activity changes

3. Improve Customer Retention Tracking

The dataset only contains single transactions per customer. The business should collect:

- Repeat purchase history
- Customer lifetime value
- Customer engagement data

This would enable accurate customer retention and forecasting analysis.